

ALGEBRA

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1. If $x + \frac{1}{x} = -13$, What is the value of $x^4 + \frac{1}{x^4}$?

यदि $x + \frac{1}{x} = -13$ है, तो $x^4 + \frac{1}{x^4}$ का मान क्या

होगा?

a) 29243 b) 28561 c) 27887 d) 27891

2. If $x + \frac{1}{x} = 5$, then the value of $x^6 + \frac{1}{x^6} = ?$

अगर $x + \frac{1}{x} = 5$ है तो $x^6 + \frac{1}{x^6} = ?$

a) 12098 b) 12048 c) 14062 d) 12092

3. If $x - \frac{1}{x} = 11$, What is the value of $(x^4 + \frac{1}{x^4})$?

यदि $x - \frac{1}{x} = 11$ है, तो $(x^4 + \frac{1}{x^4})$ का मान क्या

होगा?

a) 14163 b) 14159 c) 15127 d) 15131

4. If $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$ then the value of will be: $x^6 + \frac{1}{x^6}$

यदि $\sqrt{x} + \frac{1}{\sqrt{x}} = \sqrt{6}$ है, तो $x^6 + \frac{1}{x^6}$ का मान ज्ञात

करें।

a) 2270 b) 2502 c) 2702 d) 2712

5. If $x + \frac{1}{x} = 3, x \neq 0$, then the value of $x^7 + \frac{1}{x^7}$ is:

यदि $x + \frac{1}{x} = 3, x \neq 0$ है, तो $x^7 + \frac{1}{x^7}$ का मान

बताइए।

a) 749 b) 843 c) 746 d) 849

6. If $x + \frac{1}{x} = -3\sqrt{2}$, what is the value of $(x^5 + \frac{1}{x^5})$?

यदि $x + \frac{1}{x} = -3\sqrt{2}$, तो $(x^5 +$

$\frac{1}{x^5})$ का मान क्या होगा?

a) $-723\sqrt{2}$

b) $-720\sqrt{2}$

c) $-715\sqrt{2}$

d) $-717\sqrt{2}$

7. If $x + \frac{1}{x} = 2\sqrt{2}$, then the value of $x^7 - \frac{1}{x^7} = ?$

अगर $x + \frac{1}{x} = 2\sqrt{2}$ है तो $x^7 - \frac{1}{x^7} = ?$

a) 194 b) 26 c) 482 d) 478

8. If $x > 1$ and $x + \frac{1}{x} = 4$, then the value of $x^4 - \frac{1}{x^4} = ?$

अगर $x > 1$ और $x + \frac{1}{x} = 4$ है तो $x^4 - \frac{1}{x^4} = ?$

a) $\sqrt{3}$ b) $4\sqrt{3}$ c) $112\sqrt{3}$ d) $56\sqrt{3}$

9. If $(a + \frac{1}{a} + 3)^2 = 16$, where a is a non-zero real number, then find the value of $a^2 + \frac{1}{a^2}$.

यदि $(a + \frac{1}{a} + 3)^2 = 16$, जहाँ a एक शून्येतर वास्तविक संख्या है, तो $a^2 + \frac{1}{a^2}$ का मान ज्ञात कीजिए।

a) 3 b) 47 c) 49 d) 7

10. If $(x^2 + \frac{1}{x^2}) = 6$, and $0 < x < 1$, what is the value of $x^4 - \frac{1}{x^4}$?

यदि $(x^2 + \frac{1}{x^2}) = 6$ और $0 < x < 1$, है, तो $x^4 - \frac{1}{x^4}$ का मान क्या होगा?

a) $24\sqrt{2}$ b) $-24\sqrt{2}$ c) $-12\sqrt{10}$ d) $12\sqrt{10}$

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11. If $x^2 - 7x + 1 = 0$, and $0 < x < 1$, what is the value of $x^2 - \frac{1}{x^2}$?

यदि $x^2 - 7x + 1 = 0$, और $0 < x < 1$, है, तो $x^2 - \frac{1}{x^2}$ का मान क्या होगा?

a) $21\sqrt{5}$ b) $-21\sqrt{5}$ c) $28\sqrt{5}$ d) $-28\sqrt{5}$

12. If $x = \sqrt{3} - \sqrt{2}$, then the value of $x^3 - x^{-3}$ is:

यदि $x = \sqrt{3} - \sqrt{2}$, तो $x^3 - x^{-3}$ का मान है:

a) $22\sqrt{3}$ b) $-22\sqrt{2}$ c) $22\sqrt{2}$ d) $-22\sqrt{3}$

13. If $x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$, then the value of $x^2 + \frac{1}{x^2}$ is
 अगर $x\left(3 - \frac{2}{x}\right) = \frac{3}{x}$ है तो $x^2 + \frac{1}{x^2}$ का मान:
 a) $\frac{19}{9}$ b) $\frac{22}{9}$ c) $\frac{28}{9}$ d) $\frac{-14}{9}$

14. If $x^2 - 3x + 1 = 0$, then the value of $2\left(x^8 + \frac{1}{x^8}\right) - 5\left(x^2 + \frac{1}{x^2}\right)$ is:
 यदि $x^2 - 3x + 1 = 0$ है, तो $2\left(x^8 + \frac{1}{x^8}\right) - 5\left(x^2 + \frac{1}{x^2}\right)$ का मान ज्ञात करें।
 a) 3479 b) 4379 c) 4370 d) 4279

15. If $8x^2 + 9x + 8 = 0$, then the value of $x^3 + \frac{1}{x^3}$ is:
 यदि $8x^2 + 9x + 8 = 0$ है, तो $x^3 + \frac{1}{x^3}$ का मान ज्ञात कीजिए।
 a) $\frac{199}{212}$ b) $\frac{999}{212}$ c) $\frac{199}{512}$ d) $\frac{999}{512}$

16. If $x^2 + 1 = 3x$, then the value of $\frac{x^4 + x^{-2}}{x^2 + 5x + 1}$ is:
 यदि $x^2 + 1 = 3x$, तो $\frac{x^4 + x^{-2}}{x^2 + 5x + 1}$ का मान है:
 a) $2\frac{1}{3}$ b) $2\frac{1}{4}$ c) $4\frac{1}{2}$ d) $3\frac{1}{2}$

17. If $x^2 - 3x - 1 = 0$, then the value of $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ is:
 यदि $x^2 - 3x - 1 = 0$ तो $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ का मान है:
 a) 1 b) $\frac{11}{9}$ c) 3 d) 8

18. If $x = \sqrt{5} + 2$, then $\frac{(2x^2 - 3x - 2)}{3x^2 - 4x - 3} = ?$
 अगर $x = \sqrt{5} + 2$ है तो $\frac{(2x^2 - 3x - 2)}{3x^2 - 4x - 3} = ?$
 a) 0.1785 b) 0.525 c) 0.625 d) 0.785

19. If $x + \frac{1}{x} = 5$, then $\frac{x^4 + 3x^3 + 5x^2 + 3x + 1}{x^4 + 1} = ?$
 अगर $x + \frac{1}{x} = 5$ है तो $\frac{x^4 + 3x^3 + 5x^2 + 3x + 1}{x^4 + 1} = ?$
 a) $\frac{43}{23}$ b) $\frac{47}{21}$ c) $\frac{41}{23}$ d) $\frac{45}{21}$

20. If x is real, and $x^4 - 5x^2 - 1 = 0$, then the value of $\left(x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1\right)$ is:
 D यदि x वास्तविक है और $x^4 - 5x^2 - 1 = 0$ है, तो $\left(x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1\right)$ का मान है:
 a) 126 b) 110 c) 116 d) 96

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21. If $x > 1$ and $x^2 + \frac{1}{x^2} = 83$ then $x^3 - \frac{1}{x^3} = ?$
 अगर $x > 1$ और $x^2 + \frac{1}{x^2} = 83$ है तो $x^3 - \frac{1}{x^3} = ?$
 a) 764 b) 750 c) 756 d) 760

22. If $x^4 + \frac{1}{x^4} = 119$, and $x > 1$, then the value of $x^3 + \frac{1}{x^3} = ?$
 अगर $x^4 + \frac{1}{x^4} = 119$, और $x > 1$ है तो $x^3 + \frac{1}{x^3} = ?$
 a) 36 b) $16\sqrt{13}$ c) 18 d) $10\sqrt{13}$

23. If $x^4 + \frac{1}{x^4} = 322$, then what is the value of $x^3 - \frac{1}{x^3} = ?$
 यदि $x^4 + \frac{1}{x^4} = 322$ है, तो $x^3 - \frac{1}{x^3}$ का मान क्या है?
 a) 16 b) 96 c) 76 d) 46

24. If $x^8 - 1442x^4 + 1 = 0$, then a possible value of $x - \frac{1}{x}$ is:
 यदि $x^8 - 1442x^4 + 1 = 0$, तो $x - \frac{1}{x}$ का संभावित मान है:
 a) 5 b) 8 c) 4 d) 6

25. If $4b^2 + \frac{1}{b^2} = 2$, then the value of $8b^3 + \frac{1}{b^3} = ?$
 अगर $4b^2 + \frac{1}{b^2} = 2$ है तो $8b^3 + \frac{1}{b^3} = ?$



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- a) 0 b) 1 c) 2 d) -4

26. If $\frac{x^8+1}{x^4} = 14$, then the value of $\frac{x^{12}+1}{x^6}$ is:

यदि $\frac{x^8+1}{x^4} = 14$ है, तो $\frac{x^{12}+1}{x^6}$ का मान क्या है?

- a) 16 b) 14 c) 52 d) 64

27. If $x^2 - 12x + 33 = 0$, then what is the value of $(x-4)^2 + \left[\frac{1}{(x-4)^2}\right]$?

यदि $x^2 - 12x + 33 = 0$ है, तो $(x-4)^2 + \left[\frac{1}{(x-4)^2}\right]$ का मान क्या है?

- a) 16 b) 14 c) 18 d) 20

28. If $x^2 - 16x + 59 = 0$, then what is the value of $(x-6)^2 + \frac{1}{(x-6)^2}$?

यदि $x^2 - 16x + 59 = 0$ है, तो $(x-6)^2 + \frac{1}{(x-6)^2}$ का मान क्या है?

- a) 14 b) 18 c) 16 d) 20

29. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$, $x > 0$, then the value of $x^2(x^2 - 47)$ is .

यदि $\sqrt{x} + \frac{1}{\sqrt{x}} = 3$, $x > 0$, है, तो $x^2(x^2 - 47)$ का मान होगा:

- a) 0 b) 2 c) -1 d) 2

30. If $x^4 + x^{-4} = 47$, ($x > 0$) then the value of $(2x-3)^2$ is:

अगर $x^4 + x^{-4} = 47$, ($x > 0$) तो $(2x-3)^2$ का मान है:

- a) 2 b) 3 c) 5 d) 4

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31. If $x^2 - 9x + 1 = 0$ what is the value $x^8 - 6239x^4 + 1$?

यदि $x^2 - 9x + 1 = 0$ है, तो $x^8 - 6239x^4 + 1$ का मान क्या होगा ?

- a) 2 b) 0 c) -1 d) 1

32. If $a = 2.361$, $b = 3.263$ and $c = 5.624$, then the value of $a^3 + b^3 - c^3 + 3abc$ is

अगर $a = 2.361$, $b = 3.263$ और $c = 5.624$ है, तो $a^3 + b^3 - c^3 + 3abc$ का मान:

- a) $(a-b)(b-c)^3 + (c-a)^3$
b) $3(a-b)(b-c)(c-a)$
c) 0
d) 1

33. If $x + y + z = 0$, then $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = ?$

अगर $x + y + z = 0$ है तो $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = ?$

- a) $(xyz)^3$ b) $x^2 + y^2 + z^2$ c) 9 d) 3

34. If $a^{\frac{1}{3}} + b^{\frac{1}{3}} + c^{\frac{1}{3}} = 0$, then $(a+b+c)^6$ is equal to:

यदि $a^{\frac{1}{3}} + b^{\frac{1}{3}} + c^{\frac{1}{3}} = 0$ है, तो $(a+b+c)^6 = ?$

- a) $81abc$ b) $729a^2b^2c^2$
c) $729abc$ d) $81a^2b^2c^2$

35. If $a^3 - b^3 - c^3 = 0$ then $a^9 - b^9 - c^9 - 3a^3b^3c^3$ is

यदि $a^3 - b^3 - c^3 = 0$ है, तो $a^9 - b^9 - c^9 - 3a^3b^3c^3$ है :

- a) 1 b) 2 c) 0 d) -1

36. If $x = a(b-c)$, $y = b(c-a)$, $z = c(a-b)$ then the value of $\left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$ is:

अगर $x = a(b-c)$, $y = b(c-a)$, $z = c(a-b)$ है तो $\left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$ का मान:

- a) $\frac{xyz}{abc}$ b) 0 c) $\frac{3xyz}{abc}$ d) $\frac{2xyz}{abc}$

37. If $x = (a+b-c)$, $y = (b+c-a)$ & $z = (c+a-b)$, then $(x-a)^3 + (y-b)^3 + (z-c)^3 = ?$



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यदि $x = (a + b - c)$, $y = (b + c - a)$ और $z = (c + a - b)$ है, तो $(x - a)^3 + (y - b)^3 + (z - c)^3$ का मान ज्ञात करें।

- a) $3(a - b)(b - c)(c - a)$
 b) $3xyz$
 c) $(x - a)(y - b)(z - c)$
 d) $3abc$

38. If $x = (b - c)(a - d)$, $y = (c - a)(b - d)$, $z = (a - b)(c - d)$, then the value of $x^3 + y^3 + z^3$ is equal to:

यदि $x = (b - c)(a - d)$, $y = (c - a)(b - d)$, $z = (a - b)(c - d)$ है, तो $x^3 + y^3 + z^3$ का मान बराबर है :

- a) xyz b) $2xyz$ c) $3xyz$ d) $-3xyz$

39. Out of given responses one of the factors of $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$ is निम्न में से कौनसा $(a^2 - b^2)^3 + (b^2 - c^2)^3 + (c^2 - a^2)^3$ का एक गुणखंड है?

- a) $(a + b)(b - c)$ b) $(a + b)(a + b)$
 c) $(a - b)(a - b)$ d) $(b - c)(b - c)$

40. Simplify the following expression.

निम्न व्यंजक को सरल करें।

$$\frac{5(a^6 - b^6)^3 + 5(b^6 - c^6)^3 + 5(c^6 - a^6)^3}{2(a^3 - b^3)^3 + 2(b^3 - c^3)^3 + 2(c^3 - a^3)^3}$$

- a) $\frac{5}{2}(a^3 + b^3)(b^3 - c^3)(c^3 - a^3)$
 b) $\frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$
 c) $\frac{5}{2}(a^3 - b^3)(b^3 - c^3)(c^3 + a^3)$
 d) $\frac{5}{2}(a^3 - b^3)(b^3 + c^3)(c^3 + a^3)$

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41. The expression $(a + b - c)^3 + (a - b + c)^3 - 8a^3$ is equal to:

व्यंजक $(a + b - c)^3 + (a - b + c)^3 - 8a^3$ इसके बराबर है:

- a) $6a(a - b + c)(c - a - b)$
 b) $3a(a + b - c)(a - b + c)$
 c) $6a(a + b - c)(a - b + c)$
 d) $3a(a - b + c)(c - a - b)$

42. If $a^2 + b^2 - c^2 = 0$, then the value of $\frac{2(a^6 + b^6 - c^6)}{3a^2b^2c^2}$ is:

अगर $a^2 + b^2 - c^2 = 0$, तो $\frac{2(a^6 + b^6 - c^6)}{3a^2b^2c^2}$ का मान है:

- a) 2 b) -2 c) 1 d) 3

43. The value of $\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$ is:

$\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$ का मान पता करो?

- a) -1 b) 1 c) 3 d) -3

44. If $a^3 + b^3 + c^3 = 3abc$ and a, b, c are positive numbers, then find $\frac{2a+7b+9c}{a+2b+3c}$.

यदि $a^3 + b^3 + c^3 = 3abc$ है और a, b, c धनात्मक संख्याएँ हैं, तो $\frac{2a+7b+9c}{a+2b+3c}$ ज्ञात करो।

- a) $\frac{4}{9}$ b) 1 c) 3 d) Can't determined

45. If $x^3 + 27y^3 + 64z^3 = 36xyz$, then the relationship between x, y and z is

यदि $x^3 + 27y^3 + 64z^3 = 36xyz$ हो, तो x, y और z में सम्बन्ध है :

- a) $x + y + z = 0$ b) $x - 3y + 4z = 0$
 c) $x + 3y = 4z$ d) $x + 3y + 4z = 0$

46. If $(x - 7)^3 + (x - 8)^3 + (x + 6)^3 = 3(x - 7)(x - 8)(x + 6)$, then what is the value of x ?

यदि $(x - 7)^3 + (x - 8)^3 + (x + 6)^3 = 3(x - 7)(x - 8)(x + 6)$ है, तो x का मान क्या है ?

- a) 6 b) 8 c) 10 d) 3



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47. If $(5x+1)^3 + (x-3)^3 + 8(3x-4)^3 = 6(5x+1)(x-3)(3x-4)$, then x is equal to:
यदि $(5x+1)^3 + (x-3)^3 + 8(3x-4)^3 = 6(5x+1)(x-3)(3x-4)$ है तो x का मान निम्नलिखित के बराबर होगा:

- a) $\frac{5}{6}$ b) $\frac{1}{3}$ c) $\frac{2}{3}$ d) $\frac{3}{4}$

48. If $x + y + z = 19$, $x^2 + y^2 + z^2 = 133$ and $xz = y^2$, then the difference between z and x is:

- यदि $x + y + z = 19$, $x^2 + y^2 + z^2 = 133$ और $xz = y^2$ है, तो x और z के बीच अंतर है:
a) 6 b) 5 c) 3 d) 4

49. If $x = z = 225$ and $y = 226$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is

- अगर $x = z = 225$ और $y = 226$ तो $x^3 + y^3 + z^3 - 3xyz$ का मान:
a) 765 b) 676 c) 674 d) 576

50. If $a = 299$, $b = 298$, $c = 297$ then the value of $2a^3 + 2b^3 + 2c^3 - 6abc$ is

- यदि $a = 299$, $b = 298$, $c = 297$ हो, तो $2a^3 + 2b^3 + 2c^3 - 6abc$ का मान बताएं ?
a) 5154 b) 5267 c) 5364 d) 5456

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51. Simplify the following expression.

निम्नलिखित व्यंजक को हल कीजिए।

$$\frac{(59 \times 59 \times 59) + (54 \times 54 \times 54) + (57 \times 57 \times 57) - 3(59)(54)(57)}{(59+54+57)}$$

- a) 76 b) 170 c) 38 d) 19

52. If $x = 32.5$, $y = 34.6$ and $z = 30.9$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is $0.98k$, where K is equal to:

यदि $x = 32.5$, $y = 34.6$ और $z = 30.9$ है, तो $x^3 + y^3 + z^3 - 3xyz$ का मान $0.98k$ होता है, जहां K का मान _____ है।

- a) 1033 b) 933 c) 1026 d) 921

53. If $x + y + z = 6$ and $xy + yz + zx = 10$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

अगर $x + y + z = 6$ और $xy + yz + zx = 10$ है तो $x^3 + y^3 + z^3 - 3xyz$ का मान:
a) 36 b) 40 c) 42 d) 48

54. If $x + y + z = 1$, $xy + yz + zx = -1$, $xyz = -1$, then $x^3 + y^3 + z^3 = ?$

यदि $x + y + z = 1$, $xy + yz + zx = -1$, $xyz = -1$ है, तो $x^3 + y^3 + z^3$ का मान होगा ?

- a) 1 b) 0 c) -2 d) -1

55. If $x + y + z = 6$ and $x^2 + y^2 + z^2 = 20$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is

अगर $x + y + z = 6$ और $x^2 + y^2 + z^2 = 20$ है तो $x^3 + y^3 + z^3 - 3xyz$ का मान:
a) 64 b) 70 c) 72 d) 76

56. If $a + b + c = 15$ and $a^2 + b^2 + c^2 = 83$ then the value of $a^3 + b^3 + c^3 - 3abc$ is

अगर $a + b + c = 15$ और $a^2 + b^2 + c^2 = 83$ तो $a^3 + b^3 + c^3 - 3abc$ का मान:
a) 200 b) 180 c) 190 d) 210

57. If $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ and $x^3 + y^3 + z^3 = 3$, then what is the value of xyz ?

यदि $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ तथा $x^3 + y^3 + z^3 = 3$ है, तो xyz का मान क्या है?
a) $\frac{1}{3}$ b) $\frac{1}{6}$ c) $-\frac{1}{6}$ d) $\frac{1}{4}$



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58. If $x + y + z = 19$, $xyz = 216$ and $xy + yz + zx = 114$, then the value of $x^3 + y^3 + z^3 + xyz$ is:

यदि $x + y + z = 19$, $xyz = 216$ और $xy + yz + zx = 114$ है तो $x^3 + y^3 + z^3 + xyz$ का मान ज्ञात करें।

- a) 1225 b) 1441 c) 361 d) 577

59. If $x + y + z = 17$, $xyz = 171$ and $xy + yz + zx = 111$, then the value of $\sqrt[3]{(x^3 + y^3 + z^3 + xyz)}$ is:

यदि $x + y + z = 17$, $xyz = 171$ और $xy + yz + zx = 111$ है, तो $\sqrt[3]{(x^3 + y^3 + z^3 + xyz)}$ का मान ज्ञात करें।

- a) -64 b) 4 c) 0 d) -4

60. Simplify the following expression.

दिए गए व्यंजक का मान ज्ञात करें।

$$(2a - b - 3c)(4a^2 + b^2 + 9c^2 + 2ab + 6ac - 3bc)$$

- a) $-8a^3 + b^3 + 27c^3$
b) $8a^3 + b^3 + 27c^3$
c) $8a^3 - b^3 - 27c^3 - 18abc$
d) $8a^3 - b^3 - 27c^3 + 18abc$

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61. If $2x + 3y + 4z = 11$, $8x^3 + 27y^3 + 64z^3 = 105$ and $xyz = 1$, then the value of $4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz$ is:

यदि $2x + 3y + 4z = 11$, $8x^3 + 27y^3 + 64z^3 = 105$ और $xyz = 1$ है, तो $4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz$ का मान बताएं।

- a) 3 b) 4 c) 5 d) 6

62. If $a + \frac{1}{a} = -2$ then the value of $a^{1000} + a^{-1000}$ is

यदि $a + \frac{1}{a} = -2$ तो $a^{1000} + a^{-1000}$ का मान निकालें

- a) 2 b) 0 c) 1 d) $\frac{1}{2}$

63. If $a + \frac{1}{a} + 2 = 0$, then the value of $a^{37} - \frac{1}{a^{100}}$ is

अगर $a + \frac{1}{a} + 2 = 0$ है तो $a^{37} - \frac{1}{a^{100}}$ का मान

- a) 0 b) -2 c) 1 d) 2

64. If $\frac{a}{b} + \frac{b}{a} = 2$, then the value of $a - b = ?$

अगर $\frac{a}{b} + \frac{b}{a} = 2$ है तो $a - b = ?$

- a) 2 b) -1 c) 0 d) 1

65. If $x + \frac{1}{x+3} = -5$, then $x^2 + \frac{32}{x^2} = ?$

यदि $x + \frac{1}{x+3} = -5$ है, तो $x^2 + \frac{32}{x^2}$ का मान बराबर है :

- a) 18 b) -18 c) 14 d) -14

66. If $r + \frac{64}{r} = 16$, then the value of $r^4 + \frac{1}{r^3}$ is

यदि $r + \frac{64}{r} = 16$ है, तो $r^4 + \frac{1}{r^3}$ का मान है।

- a) 4096 b) $4096 \frac{1}{512}$
c) 512 d) $512 \frac{1}{4096}$

67. If $a + b = 10$ and $\sqrt{\frac{a}{b}} - 13 = -\sqrt{\frac{b}{a}} - 11$, then what is the value of $3ab + 4a^2 + 5b^2$?

यदि $a + b = 10$ तथा $\sqrt{\frac{a}{b}} - 13 = -\sqrt{\frac{b}{a}} - 11$ है, तो $3ab + 4a^2 + 5b^2$ का मान क्या है?

- a) 450 b) 300 c) 600 d) 750



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68. If $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} + 4 = 0$, then $x^2 + y^2 = ?$

अगर $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} + 4 = 0$ है तो $x^2 + y^2 = ?$

- a) 2 b) 4 c) -2 d) -4

69. If x is real, $x + \frac{1}{x} \neq 0$ and $x^3 + \frac{1}{x^3} = 0$, then the value of $\left(x + \frac{1}{x}\right)^4 = ?$

x एक वास्तविक संख्या है; अगर $x + \frac{1}{x} \neq 0$ और

$x^3 + \frac{1}{x^3} = 0$, तो $\left(x + \frac{1}{x}\right)^4 = ?$

- a) 4 b) 9 c) 16 d) 2

70. If $\left(x + \frac{1}{x}\right)^2 = 3$, then what is the value of $x^6 + x^{-6}$?

यदि $\left(x + \frac{1}{x}\right)^2 = 3$ है, तो $x^6 + x^{-6}$ का मान क्या है?

- a) 6 b) 2 c) -6 d) -2

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71. If $x + \frac{1}{x} = \sqrt{3}$, then $x^{18} + x^{12} + x^6 + 1 = ?$

अगर $x + \frac{1}{x} = \sqrt{3}$ है तो $x^{18} + x^{12} + x^6 + 1 = ?$

- a) 0 b) 1 c) 2 d) 3

72. If $\left(a + \frac{1}{a}\right)^2 = 3$, then the value of $a^{20} + a^{14} + a^8 + a^2 + 1 = ?$

अगर $\left(a + \frac{1}{a}\right)^2 = 3$ है तो $a^{20} + a^{14} + a^8 + a^2 + 1 = ?$

- a) 0 b) 20 c) 1 d) -1

73. If $x^3 + \frac{1}{x^3} = 0$ then the value of $x^{50} + \frac{1}{x^{50}}$ is

अगर $x^3 + \frac{1}{x^3} = 0$ है तो $x^{50} + \frac{1}{x^{50}}$ का मान बताओ।

- a) $\frac{1}{\sqrt{3}}$ b) $\sqrt{3}$ c) 1 d) 0

74. If $x + \frac{1}{x} + \sqrt{3} = 0$ then the value of $x^{17} + \frac{1}{x^{17}}$ is

अगर $x + \frac{1}{x} + \sqrt{3} = 0$ है तो $x^{17} + \frac{1}{x^{17}}$ का मान बताओ।

- a) $-\sqrt{3}$ b) $\sqrt{3}$ c) 1 d) $\frac{1}{\sqrt{3}}$

75. If $\frac{x}{y} + \frac{y}{x} = -1$ ($x, y \neq 0$), then find the value of $x^3 - y^3$.

यदि $\frac{x}{y} + \frac{y}{x} = -1$ ($x, y \neq 0$) है, तो $x^3 - y^3$ का मान ज्ञात करो।

- a) 1 b) -1 c) 0 d) 2

76. If $\frac{x}{3} + \frac{3}{x} = 1$, find the value of x^3

यदि $\frac{x}{3} + \frac{3}{x} = 1$ हो तो x^3 का मान ज्ञात करो।

- a) 1 b) -1 c) 27 d) -27

77. If $(x-3)^2 + (y-5)^2 + (z-4)^2 = 0$ then the value of $\frac{x^2}{9} + \frac{y^2}{25} + \frac{z^2}{16}$

अगर $(x-3)^2 + (y-5)^2 + (z-4)^2 = 0$ है तो $\frac{x^2}{9} + \frac{y^2}{25} + \frac{z^2}{16}$:

- a) 12 b) 9 c) 3 d) 1

78. If $(2a-1)^2 + (4b-3)^2 + (4c+5)^2 = 0$ then the value of $\frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2}$ is

अगर $(2a-1)^2 + (4b-3)^2 + (4c+5)^2 = 0$ है तो $\frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2}$:

- a) $\frac{27}{8}$ b) $\frac{19}{8}$ c) 0 d) $\frac{11}{8}$

79. If the value of $(a+b-2)^2 + (b+c-5)^2 + (c+a-5)^2 = 0$, then the value of $\sqrt{(b+c)^a + (c+a)^b - 1}$ is:

यदि $(a+b-2)^2 + (b+c-5)^2 + (c+a-5)^2 = 0$, तो $\sqrt{(b+c)^a + (c+a)^b - 1}$ का मान है:

- a) 2 b) 1 c) 3 d) 0



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80. If $x^2 + y^2 + 2x + 1 = 0$, then the value of $x^{31} + y^{35}$ is

अगर $x^2 + y^2 + 2x + 1 = 0$ हैं तो $x^{31} + y^{35}$

पता करें।

- a) -1 b) 0 c) 1 d) 2

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81. If $a^2 + b^2 = 4b + 6a - 13$, then what is the value of $a + b$?

यदि $a^2 + b^2 = 4b + 6a - 13$ है, तो $a + b$ का मान क्या है?

- a) 3 b) 2 c) 5 d) 10

82. If $2x^2 + y^2 + 6x - 2xy + 9 = 0$, then the value of $(4x^3 - y^3 + x^2y^2)$ is:

यदि $2x^2 + y^2 + 6x - 2xy + 9 = 0$ है, तो $(4x^3 - y^3 + x^2y^2)$ का मान है:

- a) 9 b) 0 c) -9 d) -3

83. $a^2 + b^2 + c^2 = 2(a + 2b + 3c) - 14$, then the value of $4a - 3b + 5c$ is

$a^2 + b^2 + c^2 = 2(a + 2b + 3c) - 14$, तो $4a - 3b + 5c$ का मान:

- a) 12 b) 13 c) 15 d) None

84. If $a^2 + b^2 + c^2 + 96 = 8(a + b - 2c)$, then $\sqrt{ab - bc + ca}$ is equal to:

यदि $a^2 + b^2 + c^2 + 96 = 8(a + b - 2c)$ तो $\sqrt{ab - bc + ca}$ निम्न में से किसके बराबर है?

- a) 6 b) $2\sqrt{2}$ c) 4 d) $2\sqrt{3}$

85. If $9a^2 + 4b^2 + c^2 + 21 = 4(3a + b - 2c)$, then the value of $(9a + 4b - c)$ is:

यदि $9a^2 + 4b^2 + c^2 + 21 = 4(3a + b - 2c)$, तो $(9a + 4b - c)$ का मान है:

- a) 16 b) 2 c) 6 d) 12

86. If $x^2 + 4y^2 + 3z^2 + \frac{19}{4} = 2\sqrt{3}(x + y + z)$, then the value of $(x - 4y + 3z)$ is:

यदि $x^2 + 4y^2 + 3z^2 + \frac{19}{4} = 2\sqrt{3}(x + y + z)$ है, तो $(x - 4y + 3z)$ का मान ज्ञात करें।

- a) $\frac{\sqrt{3}}{3}$ b) $2\sqrt{3}$ c) $\sqrt{3}$ d) $\frac{\sqrt{3}}{2}$

87. If $9a^2 + 16b^2 + c^2 + 25 = 24(a + b)$, then $(3a + 4b + 5c) = ?$

यदि $9a^2 + 16b^2 + c^2 + 25 = 24(a + b)$ है, तो $(3a + 4b + 5c) = ?$

- a) 9 b) 6 c) 7 d) 10

88. If $a^2 + b^2 + 64c^2 + 16c + 3 = 2(a + b)$, then $4a^7 + b^7 + 8c^2 = ?$

यदि $a^2 + b^2 + 64c^2 + 16c + 3 = 2(a + b)$ है, तो $4a^7 + b^7 + 8c^2 = ?$

- a) $4\frac{1}{8}$ b) $4\frac{7}{8}$ c) $5\frac{1}{8}$ d) $3\frac{7}{8}$

89. If $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$, then what is the value of $(a^3 + b^3 + c^3)$?

यदि $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$ है, तो $(a^3 + b^3 + c^3)$ का मान ज्ञात करें।

- a) -7 b) 9 c) 10 d) -4

90. The simplest form of expression $\frac{p^2 - p}{2p^3 + 6p^2} +$

$\frac{p^2 - 1}{p^2 + 3p} + \frac{p^2}{p + 1}$ को सरल करें।

- a) $2p^2$ b) $\frac{1}{2p^2}$ c) $p + 3$ d) $\frac{1}{p + 3}$

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91. If $x + y = 7$, then the value of $x^3 + y^3 + 21xy$ is

अगर $x + y = 7$ है तो $x^3 + y^3 + 21xy$ का मान:

- a) 243 b) 143 c) 343 d) 443

92. If $a + b = 1$, then $a^4 + b^4 - a^3 - b^3 - 2a^2b^2 + ab$ is equal to

यदि $a + b = 1$ है, तो $a^4 + b^4 - a^3 - b^3 - 2a^2b^2 + ab$ किसके बराबर होगा ?

- a) 1 b) 2 c) 4 d) 0

93. If $ax + by = 6$, $bx - ay = 2$ & $x^2 + y^2 = 4$ then the value of $(a^2 + b^2)$ is

यदि $ax + by = 6$, $bx - ay = 2$ और $x^2 + y^2 = 4$ है, तो $(a^2 + b^2)$ का मान है :

- a) 2 b) 4 c) 5 d) 10

94. If $[\sqrt{a^2 + b^2 + ab}] + [\sqrt{a^2 + b^2 - ab}] = 1$, then what is the value of $(1 - a^2)(1 - b^2)$?

यदि $[\sqrt{a^2 + b^2 + ab}] + [\sqrt{a^2 + b^2 - ab}] = 1$ हो, तो $(1 - a^2)(1 - b^2)$ का मान क्या होगा?

- a) $\frac{1}{4}$ b) $\frac{4}{7}$ c) $\frac{5}{4}$ d) $\frac{3}{4}$

95. If $x_1x_2x_3 = 4(4 + x_1 + x_2 + x_3)$, then what is the value of $\left[\frac{1}{2+x_1}\right] + \left[\frac{1}{2+x_2}\right] + \left[\frac{1}{2+x_3}\right]$?

यदि $x_1x_2x_3 = 4(4 + x_1 + x_2 + x_3)$ हो, तो $\left[\frac{1}{2+x_1}\right] + \left[\frac{1}{2+x_2}\right] + \left[\frac{1}{2+x_3}\right]$ का मान क्या है?

- a) 1 b) $\frac{1}{2}$ c) 2 d) $\frac{1}{3}$

96. $ab(a - b) + bc(b - c) + ca(c - a)$ is equal to:

$$ab(a - b) + bc(b - c) + ca(c - a)$$

निम्नलिखित में से किसके बराबर है?

- a) $(a + b)(b - c)(c - a)$
b) $(a - b)(b + c)(c - a)$
c) $(a - b)(b - c)(c - a)$
d) $(b - a)(b - c)(c - a)$

97. Find $ab(a + b) + bc(b + c) + ca(c + a) + 3abc$, if $a + b + c = 17$ and $ab + bc + ca = 6$?

यदि $a + b + c = 17$ और $ab + bc + ca = 6$ हो तो $ab(a + b) + bc(b + c) + ca(c + a) + 3abc$ का मान ज्ञात करें?

- a) 42 b) 119 c) 102 d) 23

98. Simplify the given expression.

दिए गए व्यंजक का मान ज्ञात करें।

$$\left(x - \frac{2}{x}\right)^3 - \left(x + \frac{2}{x}\right)^3$$

- a) $-4\left(3x + \frac{4}{x^3}\right)$ b) $-4\left(3x - \frac{4}{x^3}\right)$
c) $-4\left(x + \frac{4}{x^3}\right)$ d) $2\left(x - \frac{4}{x^3}\right)$

99. $(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$ can be expressed as:

$(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$ को कैसे लिखा जा सकता है:

- a) $(x - y)^4 - x^4$ b) $(x + y)^4 - y^4$
c) $(x + y)^4 - x^4$ d) $x^4 - (x - y)^4$

100. What is the simplified value of

$$\frac{(x+y+z)(xy+yz+zx)-xyz}{(x+y)(y+z)(z+x)} ?$$

$\frac{(x+y+z)(xy+yz+zx)-xyz}{(x+y)(y+z)(z+x)}$ का सरलीकृत मान कितना होगा?

- a) y b) z c) 1 d) x

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$\frac{(\sqrt{x+4}+\sqrt{x-4})}{(\sqrt{x+4}-\sqrt{x-4})} = \frac{7}{3}$ है तो $\frac{x+2}{x-2}$ किसके समान होगा?

- a) $\frac{17}{11}$ b) $\frac{79}{37}$ c) $\frac{11}{17}$ d) $\frac{37}{79}$

101. If $(3x + 2y)^3 + (3x - 2y)^3 = 3kx(3x^2 + 4y^2)$, then the value of k will be:

यदि $(3x + 2y)^3 + (3x - 2y)^3 = 3kx(3x^2 + 4y^2)$, है, तो k का मान ज्ञात करें।

- a) 18 b) 9 c) 3 d) 6

102. If $X + Y + Z = 0$, then what is the value of $\frac{3y^2 + x^2 + z^2}{2y^2 - xz}$?

यदि $X + Y + Z = 0$ हो, तो $\frac{3y^2 + x^2 + z^2}{2y^2 - xz}$ का मान क्या है?

- a) 2 b) 1 c) $\frac{3}{2}$ d) $\frac{5}{3}$

103. If $ab + bc + ac = 0$ then the value of $\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ac} + \frac{1}{c^2 - ab}\right)$ is

अगर $ab + bc + ac = 0$ है तो $\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ac} + \frac{1}{c^2 - ab}\right) = ?$

- a) 0 b) 1 c) 3 d) $a + b + c$

104. If $x^4 + 2x^3 + ax^2 + bx + 9$ is a perfect square, where a and b are positive real numbers, then the value of a and b are

यदि $x^4 + 2x^3 + ax^2 + bx + 9$ एक सम्पूर्ण वर्ग है जहाँ a तथा b धनात्मक वास्तविक संख्या है तो a और b का मान क्या होगा ?

- a) $a = 5, b = 6$ b) $a = 6, b = 7$
c) $a = 7, b = 6$ d) $a = 7, b = 8$

105. If $(4x + 2y)^3 + (4x - 2y)^3 = 16(Ax^3 + Bxy^2)$, then what is the value of $\frac{1}{2}\sqrt{A^2 + B^2}$?

यदि $(4x + 2y)^3 + (4x - 2y)^3 = 16(Ax^3 + Bxy^2)$, तो $\frac{1}{2}\sqrt{A^2 + B^2}$ का मान क्या है?

- a) 8 b) 3 c) 5 d) 7

106. $\frac{(\sqrt{x+4}+\sqrt{x-4})}{(\sqrt{x+4}-\sqrt{x-4})} = \frac{7}{3}$ then $\frac{x+2}{x-2}$ is equal to

107. If $A = \frac{1+2x}{1-2x}$ and $B = \frac{1-2x}{1+2x}$, then the value of $\frac{A+B}{A-B}$ is :

यदि $A = \frac{1+2x}{1-2x}$ और $B = \frac{1-2x}{1+2x}$ हो, तो $\frac{A+B}{A-B}$ का मान है :

- a) $x + \frac{1}{4x}$ b) $x - \frac{1}{4x}$ c) $\frac{1}{4x} - x$ d) $x^2 + \frac{1}{4x^2}$

108. The value of $\frac{\sqrt{0.6912} + \sqrt{0.5292}}{\sqrt{0.6912} - \sqrt{0.5292}}$ is

$\frac{\sqrt{0.6912} + \sqrt{0.5292}}{\sqrt{0.6912} - \sqrt{0.5292}}$ का मान है

- a) 1.5 b) 0.9 c) 15 d) 9

109. If

$(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$, then:

यदि

$(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$ है, तो:

- a) $\frac{a}{b} = \frac{d}{c}$ b) $\frac{a}{b} = \frac{c}{d}$ c) $\frac{a}{d} = \frac{c}{b}$ d) $\frac{b}{a} = \frac{c}{b}$

110. If $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is

अगर $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$ है तो $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ पता करें।

- a) 1 b) 2 c) 3 d) 4

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111. If $\frac{a}{1-2a} + \frac{b}{1-2b} + \frac{c}{1-2c} = \frac{1}{2}$, then $\frac{1}{1-2a} + \frac{1}{1-2b} + \frac{1}{1-2c}$ is



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अगर $\frac{a}{1-2a} + \frac{b}{1-2b} + \frac{c}{1-2c} = \frac{1}{2}$ तो $\frac{1}{1-2a} + \frac{1}{1-2b} + \frac{1}{1-2c}$ का मान:

a) -1 b) -4 c) 2 d) 4

यदि $x = \sqrt{10} + \sqrt{11}, y = \sqrt{10} - \sqrt{11}$, तो $7x^2 - 50xy + 7y^2$ का मान ज्ञात कीजिए।

a) 386 b) 1360 c) 344 d) 704

112. $\frac{m-a^2}{b^2+c^2} + \frac{m-b^2}{c^2+a^2} + \frac{m-c^2}{a^2+b^2} = 3$, then the value of m is:

$\frac{m-a^2}{b^2+c^2} + \frac{m-b^2}{c^2+a^2} + \frac{m-c^2}{a^2+b^2} = 3$ है तो m पता करें।

a) $a^2 + b^2$ b) $a^2 + b^2 + c^2$
c) $a^2 - b^2 - c^2$ d) $a^2 + b^2 - c^2$

118. If the value of $\frac{3x\sqrt{y}+2y\sqrt{x}}{3x\sqrt{y}-2y\sqrt{x}} - \frac{3x\sqrt{y}-2y\sqrt{x}}{3x\sqrt{y}+2y\sqrt{x}}$ is same as that of $\sqrt{x}\sqrt{y}$, then which of the following relations between x and y is correct?

यदि $\frac{3x\sqrt{y}+2y\sqrt{x}}{3x\sqrt{y}-2y\sqrt{x}} - \frac{3x\sqrt{y}-2y\sqrt{x}}{3x\sqrt{y}+2y\sqrt{x}}$ का मान $\sqrt{x}\sqrt{y}$ के समान है, तो x और y के बीच निम्नलिखित में से कौन सा संबंध सही है?

a) $9x - 4y = 36$ b) $9x + 4y = 24$
c) $9x + 4y = 36$ d) $9x - 4y = 24$

113. If $x^2 = y + z$, $y^2 = z + x$, $z^2 = x + y$, then the value of $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$ is

अगर $x^2 = y + z$, $y^2 = z + x$, $z^2 = x + y$ है तो $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$ का मान:

a) -1 b) 1 c) 2 d) 4

114. If $x = \frac{a}{b} + \frac{b}{a}$, $y = \frac{b}{c} + \frac{c}{b}$ and $z = \frac{c}{a} + \frac{a}{c}$, then what is the value of $xyz - x^2 - y^2 - z^2$?

यदि $x = \frac{a}{b} + \frac{b}{a}$, $y = \frac{b}{c} + \frac{c}{b}$ तथा $z = \frac{c}{a} + \frac{a}{c}$ है, तो $xyz - x^2 - y^2 - z^2$ का मान क्या है?

a) -4 b) 2 c) -1 d) -6

115. If $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ & $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$ then value of $x^2 + y^2$ is:

अगर $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ और $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$ है, तो $x^2 + y^2$:

a) 14 b) 13 c) 15 d) 10

116. If $a = \sqrt{8} - \sqrt{7}$ and $a = \frac{1}{b}$, then $\frac{a^2+b^2-3ab}{a^2+ab+b^2}$ is equal to:

अगर $a = \sqrt{8} - \sqrt{7}$ और $a = \frac{1}{b}$, तो $\frac{a^2+b^2-3ab}{a^2+ab+b^2}$ बराबर है:

a) $\frac{27}{31}$ b) $\frac{27}{32}$ c) $\frac{29}{33}$ d) $\frac{29}{31}$

117. If $x = \sqrt{10} + \sqrt{11}, y = \sqrt{10} - \sqrt{11}$, then value of $7x^2 - 50xy + 7y^2 =$ ____.

119. If $a + b = \sqrt{7}$ and $a - b = \sqrt{5}$, then find the value of $8ab(a^2 + b^2) - (a - b)^2$.

यदि $a + b = \sqrt{7}$ और $a - b = \sqrt{5}$ है, तो $8ab(a^2 + b^2) - (a - b)^2$ का मान ज्ञात कीजिए।

a) 19 b) 23 c) 21 d) 27

120. Given that x, y, z are positive real numbers, if $(x + y)^2 - z^2 = 8$, $(y + z)^2 - x^2 = 10$ and $(x + z)^2 - y^2 = 7$, then $(x + y + z)$ is equal to:

दिया गया है कि x, y, z धनात्मक वास्तविक संख्याएँ हैं, यदि $(x + y)^2 - z^2 = 8$, $(y + z)^2 - x^2 = 10$ और $(x + z)^2 - y^2 = 7$, फिर $(x + y + z)$ बराबर है:

a) 5 b) 7 c) 8 d) 6

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121. The factors of $a^2 - 1 - 2x - x^2$ are ____.

$a^2 - 1 - 2x - x^2$ के गुणखंड ____ है।



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- a) $(a - x - 1)(a - x - 1)$
 b) $(a - x + 1)(a - x - 1)$
 c) $(a + 1 + x)(a - 1 - x)$
 d) $(a - x + 1)(a - x + 1)$

122. What is the value of $1006^2 - 1007 \times 1005 + 1008 \times 1004 - 1009 \times 1003$?

- $1006^2 - 1007 \times 1005 + 1008 \times 1004 - 1009 \times 1003$ का मान क्या है?
 a) 6 b) 3 c) 12 d) 24

123. Simplify the following expression.

$$[(1 + p)(1 + p^2)(1 + p^4)(1 + p^8)(1 + p^{16})(1 - p) - 1]$$

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$$[(1 + p)(1 + p^2)(1 + p^4)(1 + p^8)(1 + p^{16})(1 - p) - 1]$$

a) $-p^{32}$ b) p^{32} c) $(1 + p^{32})$ d) $(1 - p^{32})$

124. $(2 + 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1)(2^{32} + 1)(2^{64} + 1)$ is:

- a) $2^{256} - 1$ b) $2^{256} + 1$
 c) $2^{128} - 1$ d) $2^{128} + 1$

125. What is the simplified value of $(x^{128} + 1)(x^{32} + 1)(x^{64} + 1)(x^{16} + 1)(x^8 + 1)(x^4 + 1)(x^2 + 1)(x + 1)$?
 $(x^{128} + 1)(x^{32} + 1)(x^{64} + 1)(x^{16} + 1)(x^8 + 1)(x^4 + 1)(x^2 + 1)(x + 1)$ का सरलीकृत मान क्या है ?

- a) $x^{256} - 1$ b) $\frac{x^{128}-1}{x-1}$ c) $\frac{x^{64}-1}{x-1}$ d) $\frac{x^{256}-1}{x-1}$

126. A complete factorization of $x^4 + 64$ is

$x^4 + 64$ का सम्पूर्ण गुणखंडन करें :

- a) $(x^2 + 8)^2$
 b) $(x^2 + 8)(x^2 - 8)$
 c) $(x^2 - 4x + 8)(x^2 - 4x - 8)$
 d) $(x^2 + 4x + 8)(x^2 - 4x + 8)$

127. The term, that should be added to $(4x^2 + 8x)$ so that resulting expression be a perfect square is:

$(4x^2 + 8x)$ में क्या जोड़ा जाए ताकि ये पूर्ण वर्ग बन जाए?

- a) $2x$ b) 2 c) 1 d) 4

128. The expression $x^4 - x^2 + k$ a perfect square if the value of k is

K के किस मान के लिए $x^4 - x^2 + k$ एक पूर्ण वर्ग होगा?

- a) 1 b) 0 c) $\frac{1}{4}$ d) $-\frac{1}{4}$

129. For what value(s) of k will the expression $p + \frac{1}{9}\sqrt{p} + k^2$ be a perfect square

k के किस मान/किन मानों के लिए व्यंजक $p + \frac{1}{9}\sqrt{p} + k^2$ एक पूर्ण वर्ग होगा?

- a) $k = \pm \frac{1}{8}$ b) $k = \pm \frac{1}{9}$
 c) $k = \pm \frac{1}{21}$ d) $k = \pm \frac{1}{18}$

130. If $N = (12345)^2 + 12345 + 12346$, then what is the value of $\sqrt{N}$?

यदि $N = (12345)^2 + 12345 + 12346$ है, तो $\sqrt{N}$ का मान क्या है?

- a) 12346 b) 12345 c) 12344 d) 12347

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131. If the expression $x + 809436 \times 809438$ be a perfect square, then the value of x is

अगर $x + 809436 \times 809438$ एक पूर्ण वर्ग है, तो x का मान पता करो।

- a) 809436 b) 809438 c) 0 d) 1



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132. If $5x + 9y = 5$ and $125x^3 + 729y^3 = 120$, then the value of the product of x and y is

अगर $5x + 9y = 5$ और $125x^3 + 729y^3 = 120$ तो x और y का गुणनफल क्या होगा?

- a) 45 b) $\frac{1}{9}$ c) $\frac{1}{135}$ d) 135

133. If $p = 999$, then the value of $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ is

अगर $p = 999$, तो $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ का मान:

- a) 1000 b) 999 c) 998 d) 1002

134. If $p^3 + 3p^2 + 3p = 26$, then the value of $p^2 + 2p$ is:

यदि $p^3 + 3p^2 + 3p = 26$ है, तो $p^2 + 2p$ का मान ज्ञात कीजिए।

- a) 8 b) 12 c) 10 d) 15

135. If $(8x^3 + 27y^3) \div (2x + 3y) = (Ax^2 + Bxy + Cy^2)$, then the value of $(5A + 4B + 3C)$ is

यदि $(8x^3 + 27y^3) \div (2x + 3y) = (Ax^2 + Bxy + Cy^2)$ तो $(5A + 4B + 3C)$ का मान है:

- a) 27 b) 24 c) 23 d) 71

136. If $8x^3 - 27y^3 = (Ax + By)(Cx^2 - Dy^2 + 6xy)$, then $(A + B + C - D)$ is equal to:

यदि $8x^3 - 27y^3 = (Ax + By)(Cx^2 - Dy^2 + 6xy)$ है, तो $(A + B + C - D)$ निम्नलिखित में से किसके बराबर है?

- a) -12 b) 12 c) 15 d) 9

137. If $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)(Ax^2 + By^2 + Cxy)$, then the value of $A^2 + B^2 - C^2$ is:

यदि $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)(Ax^2 + By^2 + Cxy)$ है, तो $A^2 + B^2 - C^2$ का क्या मान है:

- a) 11 b) 7 c) 19 d) 10

138. If $24\sqrt{3}x^3 + 5\sqrt{5}y^3 = (2\sqrt{3}x + \sqrt{5}y) \times (Ax^2 + Bxy + Cy^2)$, then what is the value of $(A^2 - B^2 + C^2)$?

यदि $24\sqrt{3}x^3 + 5\sqrt{5}y^3 = (2\sqrt{3}x + \sqrt{5}y) \times (Ax^2 + Bxy + Cy^2)$ है, तो $(A^2 - B^2 + C^2)$ का मान क्या होगा ?

- a) 108 b) 128 c) 109 d) 139

139. If $x^6 - 512y^6 = (x^2 + Ay^2)(x^4 - Bx^2y^2 + Cy^4)$, then what is the value of $(A + B - C)$?

अगर $x^6 - 512y^6 = (x^2 + Ay^2)(x^4 - Bx^2y^2 + Cy^4)$, तो $(A + B - C)$ का मान क्या है?

- a) -72 b) 72 c) -80 d) 48

140. If $[8(x + y)^3 - 27(x - y)^3] \div (5y - x) = Ax^2 + Bxy + Cy^2$ then the value of $(A + B + C)$ is:

यदि $[8(x + y)^3 - 27(x - y)^3] \div (5y - x) = Ax^2 + Bxy + Cy^2$ तो $(A + B + C)$ का मान है:

- a) 27 b) 24 c) 16 d) 18

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141. If $8(a + b)^3 + (a - b)^3 = (3a + b)(Aa^2 + Bab + Cb^2)$, then what is the value of $(A + B - C)$?

यदि $8(a + b)^3 + (a - b)^3 = (3a + b)(Aa^2 + Bab + Cb^2)$ है, तो $(A + B - C)$ का मान क्या होगा ?

- a) 2 b) 4 c) 10 d) 11

142. If $a^3 + b^3 = 110$ and $a + b = 5$, then $(a + b)^2 - 3ab$ is equal to



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अगर $a^3 + b^3 = 110$ और $a + b = 5$, तो

$(a + b)^2 - 3ab$ बराबर है

- a) 52 b) 32 c) 42 d) 22

- a) 100 b) 10000 c) 100000 d) 10

148. $\frac{(253)^3 + (247)^3}{25.3 \times 2.53 - 62.491 + 2.47 \times 24.7} = 50 \times 10^k, k = ?$
a) 3 b) 4 c) 2 d) -3

143. On simplification, $\frac{x^3 - y^3}{x[(x+y)^2 - 3xy]} \div$

$\frac{y[(x-y)^2 + 3xy]}{x^3 + y^3} \times \frac{(x+y)^2 - (x-y)^2}{x^2 - y^2}$ is equal to:

सरलीकरण पर, $\frac{x^3 - y^3}{x[(x+y)^2 - 3xy]} \div \frac{y[(x-y)^2 + 3xy]}{x^3 + y^3} \times \frac{(x+y)^2 - (x-y)^2}{x^2 - y^2}$ इसके बराबर है:

- a) 4 b) 1 c) $\frac{1}{2}$ d) $\frac{1}{4}$

149. Simplify the following.

$\frac{0.01 \times 0.01 \times 0.01 + 0.003 \times 0.003 \times 0.003}{0.05 \times 0.05 - 0.015 \times 0.05 + 0.015 \times 0.015}$

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कीजिए। $\frac{0.01 \times 0.01 \times 0.01 + 0.003 \times 0.003 \times 0.003}{0.05 \times 0.05 - 0.015 \times 0.05 + 0.015 \times 0.015}$

- a) $\frac{13}{25} \times 10^3$ b) $\frac{13}{15} \times 10^{-3}$
c) $\frac{13}{15} \times 10^3$ d) $\frac{13}{25} \times 10^{-3}$

144. If $P = \frac{x^3 + y^3}{(x-y)^2 + 3xy}$, $Q = \frac{(x+y)^2 - 3xy}{x^3 - y^3}$ and $R = \frac{(x+y)^2 + (x-y)^2}{x^2 - y^2}$, then what is the value of $(P \div Q) \times R$?

अगर $P = \frac{x^3 + y^3}{(x-y)^2 + 3xy}$, $Q = \frac{(x+y)^2 - 3xy}{x^3 - y^3}$ और $R = \frac{(x+y)^2 + (x-y)^2}{x^2 - y^2}$, तो $(P \div Q) \times R$

का मान क्या है?

- a) $2(x^2 + y^2)$ b) $4xy$
c) $x^2 + y^2$ d) $2xy$

150. If $a^2 + b^2 + ab = 0$, then $(a^3 - b^3)$ is equal to

अगर $a^2 + b^2 + ab = 0$ है, तो $(a^3 - b^3)$:

- a) 0 b) 1 c) $(a + b)^3$ d) $a^2b^4 + a^4b^2$

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145. If $P = \frac{x^4 - 8x}{x^3 - x^2 - 2x}$, $Q = \frac{x^2 + 2x + 1}{x^2 - 4x - 5}$ and $R = \frac{2x^2 + 4x + 8}{x - 5}$, then $(P \times Q) \div R$ is equal to:

अगर $P = \frac{x^4 - 8x}{x^3 - x^2 - 2x}$, $Q = \frac{x^2 + 2x + 1}{x^2 - 4x - 5}$ और $R = \frac{2x^2 + 4x + 8}{x - 5}$ तो $(P \times Q) \div R$ बराबर है:

- a) $\frac{1}{2}$ b) 1 c) 2 d) 4

151. If $\frac{a}{b} + \frac{b}{a} = 1$, $a \neq 0, b \neq 0$ the value of $a^3 + b^3$ is

अगर $\frac{a}{b} + \frac{b}{a} = 1$, $a \neq 0, b \neq 0$ तो $a^3 + b^3$ का मान:

- a) 0 b) 1 c) -1 d) 2

146. Simplify the following expression.

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$\frac{(62 \times 62 \times 62) - 3(62 \times 62 \times 22) + 3(62 \times 22 \times 22) - (22 \times 22 \times 22)}{8 \times 8 \times 8}$

- a) 225 b) 1250 c) 125 d) 25

147. $\frac{775 \times 775 \times 775 + 225 \times 225 \times 225}{77.5 \times 77.5 + 22.5 \times 22.5 - 77.5 \times 22.5}$ is equal to:

$\frac{775 \times 775 \times 775 + 225 \times 225 \times 225}{77.5 \times 77.5 + 22.5 \times 22.5 - 77.5 \times 22.5}$ का मान ज्ञात करें।

152. If $a = \frac{b^2}{b-a}$, then the value of $a^3 + b^3$ is

अगर $a = \frac{b^2}{b-a}$, तो $a^3 + b^3$ का मान:

- a) 6ab b) 0 c) 1 d) 2

153. If $\frac{1}{x+y} = \frac{1}{x} + \frac{1}{y}$ then the value of $x^3 - y^3$ is

अगर $\frac{1}{x+y} = \frac{1}{x} + \frac{1}{y}$ तो $x^3 - y^3$ का मान:

- a) 0 b) 1 c) -1 d) 2

154. If $a^4 + b^4 = a^2b^2$, then $(a^6 + b^6)$ equals



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अगर $a^4 + b^4 = a^2 b^2$ है, तो $(a^6 + b^6)$:

- a) 0 b) 1
c) $a^2 + b^2$ d) $a^2 b^4 + a^4 b^2$

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155. If $X + Y = 10$ and $XY = 4$, then what is the value of $x^4 + y^4$?

यदि $X + Y = 10$ तथा $xy = 4$ है, तो $x^4 + y^4$

का मान क्या है?

- a) 8464 b) 8432 c) 7478 d) 6218

156. If $a + b = 27$ and $a^3 + b^3 = 5427$, then find ab.

यदि $a + b = 27$ और $a^3 + b^3 = 5427$ है, तो ab का मान ज्ञात करें।

- a) 143 b) 135 c) 176 d) 149

157. If $x + y = 7$ and $xy = 10$, then the value of $\left(\frac{1}{x^3} + \frac{1}{y^3}\right)$ is:

यदि $x + y = 7$ और $xy = 10$, तो $\left(\frac{1}{x^3} + \frac{1}{y^3}\right)$ का मान है:

- a) 0.543 b) 0.131 c) 0.133 d) 0.454

158. If $x - y = 4$, and $x^3 - y^3 = 316$, then the value of $x^4 + y^4$ is:

यदि $x - y = 4$ और $x^3 - y^3 = 316$ है, तो $x^4 + y^4$ का मान ज्ञात करें।

- a) 2248 b) 2482 c) 2428 d) 2284

159. If $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ and $y = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ then the value of $x^3 + y^3$ is

अगर $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ और $y = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ तो $x^3 + y^3$ का मान:

- a) 950 b) 730 c) 650 d) 970

160. $\left(\frac{\sqrt{5}+1}{\sqrt{5}-1}\right)^3 + \left(\frac{\sqrt{5}-1}{\sqrt{5}+1}\right)^3 = ?$

- a) 18 b) 36 c) 27 d) 16

161. If $x = \sqrt{5} + 1$ & $y = \sqrt{5} - 1$ then the value of $\frac{x^2}{y^2} + \frac{y^2}{x^2} + 4\left[\frac{x}{y} + \frac{y}{x}\right] + 6$ is

यदि $x = \sqrt{5} + 1$ तथा $y = \sqrt{5} - 1$ है, तो $\frac{x^2}{y^2} + \frac{y^2}{x^2} + 4\left[\frac{x}{y} + \frac{y}{x}\right] + 6$ का मान क्या है?

- a) 31 b) $23\sqrt{5}$ c) $27\sqrt{5}$ d) 25

162. If $x = \sqrt{3} - \frac{1}{\sqrt{3}}$ and $y = \sqrt{3} + \frac{1}{\sqrt{3}}$ then the value of $\frac{x^2}{y} + \frac{y^2}{x}$ is

अगर $x = \sqrt{3} - \frac{1}{\sqrt{3}}$ और $y = \sqrt{3} + \frac{1}{\sqrt{3}}$ तो $\frac{x^2}{y} + \frac{y^2}{x}$ का मान:

- a) $\sqrt{3}$ b) $3\sqrt{3}$ c) $16\sqrt{3}$ d) $2\sqrt{3}$

163. The difference between two numbers is 3 and the difference between their cubes is 999. Find the difference between their squares.

दो संख्याओं के बीच का अंतर 3 है और उनके घनों के बीच का अंतर 999 है। उनके वर्गों का अंतर ज्ञात कीजिए।

- a) 81 b) 63 c) 36 d) 18

164. If $a^2 + b^2 = 99$ and $ab = 11$, ($a > 0, b > 0$) then the value of $(a^3 + b^3)$ is :

यदि $a^2 + b^2 = 99$ और $ab = 11$, ($a > 0, b > 0$) तो $(a^3 + b^3)$ का मान है:

- a) 1250 b) 968 c) 1100 d) 1080

165. If $a^2 + b^2 = 88$ and $ab = 6$, ($a > 0, b > 0$) then what is the value of $(a^3 + b^3)$ is ?

यदि $a^2 + b^2 = 88$ और $ab = 6$, ($a > 0, b > 0$) तो $(a^3 + b^3)$ का मान क्या है?

- a) 980 b) 1180 c) 820 d) 1000

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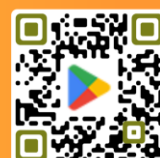
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166. If $x^3 + y^3 = 416$ and $x + y = 8$, then find $x^4 + y^4$.

यदि $x^3 + y^3 = 416$ और $x + y = 8$ है तो $x^4 + y^4$ का मान ज्ञात कीजिए।

a) 3002 b) 3204 c) 3004 d) 3104

167. x, y are two positive numbers such that $x > y$. If $x^4 + y^4 = 706$ and $xy = 15$, then the value of $2x + 3y$ is:

x, y दो ऐसी धनात्मक संख्याएँ हैं कि $x > y$ है।

यदि $x^4 + y^4 = 706$ और $xy = 15$ है, तो $2x + 3y$ का मान ज्ञात करें।

a) 19 b) 20 c) 18 d) 15

168. If $8a^3 + b^3 = 16$ and $2a + b = 4$, then find the value of $16a^4 + b^4$.

यदि $8a^3 + b^3 = 16$ और $2a + b = 4$ है, तो $16a^4 + b^4$ का मान क्या होगा?

a) 32 b) 36 c) 28 d) 38

169. If $16x^2 + y^2 = 48$ and $xy = 2$, $x, y > 0$, then the value of $(64x^3 + y^3)$ is:

यदि $16x^2 + y^2 = 48$ और $xy = 2$, $x, y > 0$ है, तो $(64x^3 + y^3)$ का मान ज्ञात करें।

a) 320 b) 300 c) 240 d) 340

170. If $\left(x^2 + \frac{1}{49x^2}\right) = 15\frac{5}{7}$, then what is the value of $\left(x + \frac{1}{7x}\right)$?

यदि $\left(x^2 + \frac{1}{49x^2}\right) = 15\frac{5}{7}$ है, तो $\left(x + \frac{1}{7x}\right)$ का मान क्या होगा?

a) 7 b) ± 7 c) ± 4 d) 4

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171. If $x + \frac{1}{15x} = 3$ then the value of $9x^3 + \frac{1}{375x^3}$ will be:

यदि $x + \frac{1}{15x} = 3$ है, तो $9x^3 + \frac{1}{375x^3}$ का मान ज्ञात करें।

a) 237.6 b) 376.2 c) 273.6 d) 367.2

172. If $x + \frac{1}{16x} = 3$, then the value of $16x^3 + \frac{1}{256x^3}$ is:

यदि $x + \frac{1}{16x} = 3$ है, तो $16x^3 + \frac{1}{256x^3}$ का मान होगा:

a) 423 b) 441 c) 432 d) 414

173. If $x^4 + x^2y^2 + y^4 = 273$ and $x^2 - xy + y^2 = 13$ then the value of xy is:

अगर $x^4 + x^2y^2 + y^4 = 273$ और $x^2 - xy + y^2 = 13$ है तो xy का मान क्या होगा?

a) 4 b) 10 c) 6 d) 8

174. If $x^4 + x^2y^2 + y^4 = 21$ and $x^2 + xy + y^2 = 7$ then the value of $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)$ is:

अगर $x^4 + x^2y^2 + y^4 = 21$ और $x^2 + xy + y^2 = 7$ है तो $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)$ का मान बताओ।

a) $\frac{5}{4}$ b) $\frac{7}{4}$ c) $\frac{7}{3}$ d) $\frac{5}{2}$

175. If $x^4 + y^4 + x^2y^2 = 117$ and $x^2 + y^2 - xy = 3(4 + \sqrt{3})$, then the value of $(x^2 + y^2)$ will be:

यदि $x^4 + y^4 + x^2y^2 = 117$ और $x^2 + y^2 - xy = 3(4 + \sqrt{3})$ है, $(x^2 + y^2)$ का ज्ञात करें।

a) 9 b) $6\sqrt{3}$ c) 12 d) $13\sqrt{3}$

176. If $x^4 + x^2y^2 + y^4 = 35$, $x^2 + xy + y^2 = 7$ and $x, y > 0$ then the value of $x^3 + y^3$ is:

अगर $x^4 + x^2y^2 + y^4 = 35$, $x^2 + xy + y^2 = 7$ और $x, y > 0$ है तो $x^3 + y^3$ का मान बताओ।

a) $10\sqrt{2}$ b) $22\sqrt{2}$ c) $16\sqrt{2}$ d) Can't say



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177. If $x^8 + x^4y^4 + y^8 = 144$ and $x^4 + x^2y^2 + y^4 = 18$, then what is the value of $x^2 + y^2$?
 अगर $x^8 + x^4y^4 + y^8 = 144$ और $x^4 + x^2y^2 + y^4 = 18$, है तो $x^2 + y^2$ का मान बताओ.
 a) 23 b) $\sqrt{23}$ c) 5 d) $\sqrt{18}$

178. If $x^8 + x^4y^4 + y^8 = 273$, $x^4 + x^2y^2 + y^4 = 21$ and $x, y > 0$, then what is the value of $(x + y)$?
 अगर $x^8 + x^4y^4 + y^8 = 273$ और $x^4 + x^2y^2 + y^4 = 21$, $(x, y > 0)$ है तो $(x + y)$ का मान बताओ.
 a) 6 b) 12 c) 3 d) Can't say

179. If $x^2 + xy + y^2 = 21$ and $x + \sqrt{xy} + y = 7$, then the value of $x - y$ is
 अगर $x^2 + xy + y^2 = 21$ और $x + \sqrt{xy} + y = 7$ है तो $x - y$ का मान बताओ.
 a) 9 b) 3 c) 5 d) Can't say

180. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and $x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of $(x - y)$ is:
 यदि $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ और $x^2 - xy + y^2 = 5\frac{1}{4}$ है, तो $(x - y)$ का मान बताइए।
 a) $\frac{5}{2}$ b) $\frac{3}{4}$ c) $\frac{5}{4}$ d) $\frac{3}{2}$

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181. If $16a^4 + 36a^2b^2 + 81b^4 = 91$ and $4a^2 + 9b^2 - 6ab = 13$, then what is the value of $3ab$?
 अगर $16a^4 + 36a^2b^2 + 81b^4 = 91$ और $4a^2 + 9b^2 - 6ab = 13$ है तो $3ab$ का मान बताओ.
 a) -3 b) 5 c) $\frac{3}{2}$ d) $-\frac{3}{2}$

182. If

$$1 + 4x^2 + 16x^4 = 512, 1 - 2x + 4x^2 = 64$$

, then the value of $1 + 2x + 4x^2$ is:

यदि $1 + 4x^2 + 16x^4 = 512$ है और $1 - 2x + 4x^2 = 64$ है, तो $1 + 2x + 4x^2$ का

मान कितना होगा?

- a) 6 b) 8 c) 12 d) 10

Answer key

1. C	2. A	3. C	4. C	5. B
6. C	7. D	8. C	9. B	10. B
11. B	12. B	13. B	14. B	15. D
16. B	17. A	18. C	19. A	20. A
21. C	22. D	23. C	24. D	25. A
26. C	27. B	28. B	29. C	30. C
31. B	32. C	33. D	34. B	35. C
36. C	37. A	38. C	39. A	40. B
41. A	42. B	43. A	44. C	45. D
46. D	47. A	48. B	49. B	50. C
51. D	52. A	53. A	54. A	55. C
56. B	57. B	58. A	59. D	60. C
61. A	62. A	63. B	64. C	65. A
66. B	67. B	68. C	69. B	70. D
71. A	72. C	73. C	74. B	75. C
76. D	77. C	78. C	79. C	80. A
81. C	82. B	83. B	84. C	85. D
86. C	87. C	88. C	89. A	90. B
91. C	92. D	93. D	94. D	95. B
96. D	97. C	98. A	99. D	100. C
101. D	102. A	103. A	104. C	105. C
106. B	107. A	108. C	109. B	110. D
111. D	112. B	113. B	114. A	115. A
116. A	117. C	118. D	119. A	120. A
121. C	122. A	123. A	124. C	125. D
126. D	127. D	128. C	129. D	130. A



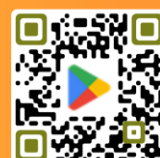
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131. D	132. C	133. A	134. A	135. C
136. B	137. B	138. C	139. C	140. C
141. A	142. D	143. A	144. A	145. A
146. C	147. C	148. B	149. D	150. A
151. A	152. B	153. A	154. A	155. B
156. C	157. C	158. B	159. D	160. A
161. D	162. B	163. B	164. B	165. C
166. D	167. A	168. A	169. A	170. C
171. A	172. A	173. A	174. A	175. C
176. A	177. B	178. C	179. B	180. A
181. D	182. B			

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